Assignment 4

Q1. Docker Compose: Nginx Static Website

Question

Create a Docker Compose setup to run an Nginx container that serves a custom HTML page.

Requirements

1. Create a folder named website
2. Create a file index.html with content:

   # Hello from Docker Compose

3. Use Docker Compose to:
   - Run an Nginx container
   - Map port 8080 to 80
   - Mount website folder to /usr/share/nginx/html

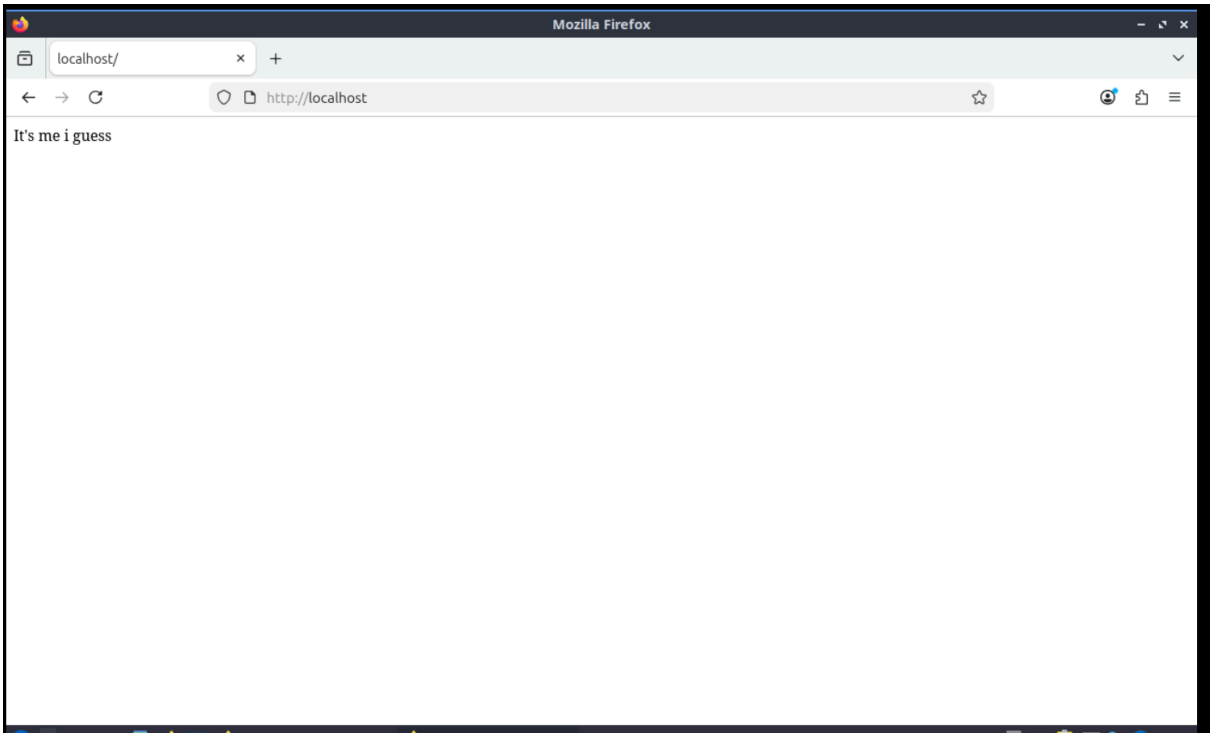
   ---

4. Open browser and verify output

```
mojorojo@mojorojo-vmware:~/website$ vim index.html
mojorojo@mojorojo-vmware:~/website$ vim docker-compose.yaml
mojorojo@mojorojo-vmware:~/website$ vim Dockerfile
mojorojo@mojorojo-vmware:~/website$ vim docker-compose.yaml
mojorojo@mojorojo-vmware:~/website$ cat Dockerfile
FROM nginx
WORKDIR /app
COPY index.html /app
mojorojo@mojorojo-vmware:~/website$ vim DO
mojorojo@mojorojo-vmware:~/website$ vim Dockerfile
mojorojo@mojorojo-vmware:~/website$ docker compose up
[+] Building 0.7s (9/9) FINISHED
 => [internal] load local bake definitions                                                                    0.0s
 => => reading from stdin 512B                                                                                0.0s
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 86B                                                                           0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                               0.0s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 83B                                                                              0.0s
 => CACHED [1/2] FROM docker.io/library/nginx:latest@sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b  0.1s
 => => resolve docker.io/library/nginx:latest@sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0    0.1s
 => [2/2] COPY index.html /usr/share/nginx/html                                                               0.0s
 => exporting to image                                                                                        0.2s
```

```
mojorojo@mojorojo-vmware:~/website$ docker ps
CONTAINER ID   IMAGE                                  COMMAND                  CREATED          STATUS          PORTS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
NAMES
15ef7f21cb27   website-custom_web                   "/docker-entrypoint.…"   9 seconds ago    Up 8 seconds    80/tcp
custom_web_container
9f026da6d396   gcr.io/k8s-minikube/kicbase:v0.0.50   "/usr/local/bin/entr…"   30 minutes ago   Up 30 minutes   127.0.0.1:32768->22/tcp, 127.0.0.1
:32769->2376/tcp, 127.0.0.1:32770->5000/tcp, 127.0.0.1:32771->8443/tcp, 127.0.0.1:32772->32443/tcp   minikube
```

LOCALHOST



docker-compose.yaml

```
mojorojo@mojorojo-vmware:~$ cat website/docker-compose.yaml
services:
  custom_web:
    build:  .
    container_name:  custom_web_container
```

Dockerfile

```
mojorojo@mojorojo-vmware:~$ cat website/Dockerfile
FROM nginx
COPY index.html /usr/share/nginx/html
```

Q2. Docker Compose: Python Script Container

Question

Run a simple Python script using Docker Compose.

Requirements

1. Create a file app.py: print("Hello from Python Container")

```
mojorojo@mojorojo-vmware:~/py-compose$ cat app.py
inp = int(input("ENter the number between 1-2"))
if inp == 1:
    print("You ENterd 1")
else:
    print("YOu enter number other than 1")
mojorojo@mojorojo-vmware:~/py-compose$
```

2. Create a Dockerfile to run this script

```
mojorojo@mojorojo-vmware:~/py-compose$ cat Dockerfile
FROM python:3.11-alpine3.22
WORKDIR /app
COPY  app.py /app
CMD ["python","app.py"]
mojorojo@mojorojo-vmware:~/py-compose$
```

3. Use Docker Compose to run this container

```
mojorojo@mojorojo-vmware:~/py-compose$ cat docker-compose.yaml
services:
  py_hello:
    build:  .
    container_name:  py_helllo
    command: ["python","app.py"]
    stdin_open: true  # Corresponds to -i
    tty: true         # Corresponds to -t
mojorojo@mojorojo-vmware:~/py-compose$
```

4. Container should execute script and exit

```
mojorojo@mojorojo-vmware:~/py-compose$ docker compose run py_hello
Container py-compose-py_hello-run-3db36993e065 Creating
Container py-compose-py_hello-run-3db36993e065 Created
ENter the number between 1-269
YOu enter number other than 1
```

Q3. Docker Compose: MySQL with Volume

Question

Create a MySQL container using Docker Compose with persistent storage.

Requirements

1. Use MySQL image

docker-compose.yml

```
mojorojo@mojorojo-vmware:~/composemysql$ cat docker-compose.yml
services:
  my_sql:
    image: mysql:8.0          # Use the official image unless you have a custom Dockerfile
    container_name: my-sql-container
    ports:
      - "3306:3306"           # Allows you to connect from your host machine
    environment:
      MYSQL_ROOT_PASSWORD: acts
    volumes:
      - dockerv1:/var/lib/mysql # Maps the named volume to the MySQL data directory

volumes:
  dockerv1:
```

Dockerfile

```
mojorojo@mojorojo-vmware:~/composemysql$ cat Dockerfile
FROM    mysql:lts-oracle
WORKDIR /mysql-app
```

1. Set:
    - root password
    - database name

```
mojorojo@mojorojo-vmware:~/composemysql$ docker exec -it my-sql-container mysql -u root -pacts
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> create database mydb;
Query OK, 1 row affected (0.01 sec)

mysql> use mydb;
Database changed
mysql> create table emp
    -> id
    -> name
    -> exit
    -> ^C
mysql> use table
ERROR 1049 (42000): Unknown database 'table'
mysql> use emp
ERROR 1049 (42000): Unknown database 'emp'
mysql> create table if not exists student(id int primary key auto_increment, name varchar(100), score int);
Query OK, 0 rows affected (0.02 sec)

mysql> insert into student (name,score) values ('abhay',001);
Query OK, 1 row affected (0.01 sec)
```

2. Create a named volume
3. Start container and:
   - Enter MySQL using docker exec
   - Create table test
4. Restart container and verify data still exists

```
mojorojo@mojorojo-vmware:~/composemysql$ docker compose restart
[+] restart 0/1
 ⠿ Container my-sql-container Restarting
```

```
mojorojo@mojorojo-vmware:~/composemysql$ docker exec -it my-sql-container mysql -u root -pacts
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.45 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> use mydb
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> select*from student;
+----+-------+-------+
| id | name  | score |
+----+-------+-------+
|  1 | abhay |     1 |
+----+-------+-------+
1 row in set (0.00 sec)

mysql> exit
Bye
```

Q4. Docker Swarm: Nginx Service

Question

Deploy Nginx service in Docker Swarm.

Requirements

1. Initialize swarm: docker swarm init

```
mojorojo@mojorojo-vmware:~/pythonswarm$ docker swarm init
Swarm initialized: current node (ntopelqu76873bde7nyce2qya) is now a manager.

To add a worker to this swarm, run the following command:

    docker swarm join --token SWMTKN-1-26hgh67ml2zjstj55w5yebw7mhl6iqub8t6w4rh1m4bqh9dkr6-apv17cv1hg0eoffd9serw4rm2 192.168.81.148:2377

To add a manager to this swarm, run 'docker swarm join-token manager' and follow the instructions.
```

1. Create service:
    - Name nginx-service
    - Image nginx
    - Replicas 2

```
mojorojo@mojorojo-vmware:~/nginxswarm$ docker build -t nginx .
[+] Building 0.4s (6/6) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 74B                                                                                       0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                                           0.0s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [1/2] FROM docker.io/library/nginx:latest@sha256:07970dac3f2466efc05bf19171cac99e6e98dc9a2dcacf53c994ac5a66e02e8f      0.1s
 => => resolve docker.io/library/nginx:latest@sha256:07970dac3f2466efc05bf19171cac99e6e98dc9a2dcacf53c994ac5a66e02e8f      0.0s
 => [2/2] WORKDIR /APP                                                                                                    0.0s
 => exporting to image                                                                                                    0.1s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:4d11307820a31244b77d451f96d5f952669455723f6fcf2472d8bfe7ebb8f8b9                         0.0s
 => => exporting config sha256:8bb6ad7a49b6d3a6a39a3ad51e928af2e22b5cc95957d6ea3e0945b43cffb2b6                           0.0s
 => => exporting attestation manifest sha256:1218f4cc92f24c008129ad5299576d659830cad730c8d22179d5971736d691dd             0.0s
 => => exporting manifest list sha256:7977d376bf700f7e200aaa32a3bf673ef62c1efa284ac7fef169274049a5c57a                    0.0s
 => => naming to docker.io/library/nginx:latest                                                                           0.0s
 => => unpacking to docker.io/library/nginx:latest                                                                        0.0s
```

```
mojorojo@mojorojo-vmware:~/nginxswarm$ docker service create --name nginx-service --replicas 2 nginx
ia6tjjmldfpt66wb4m8aavxtd
overall progress: 2 out of 2 tasks
1/2: running   [==================================================>]
2/2: running   [==================================================>]
verify: Service ia6tjjmldfpt66wb4m8aavxtd converged
mojorojo@mojorojo-vmware:~/nginxswarm$
```

1. Expose port 8080

```
mojorojo@mojorojo-vmware:~/nginxswarm$ cat Dockerfile
FROM    nginx
WORKDIR /app
EXPOSE  8080
```

1. Verify using: docker service ls docker service ps nginx-service

```
mojorojo@mojorojo-vmware:~/nginxswarm$ docker service ls
ID             NAME             MODE         REPLICAS   IMAGE          PORTS
ia6tjjmldfpt   nginx-service    replicated   2/2        nginx:latest
iah7c0x0ad8o   python-service   replicated   3/3        nginx:latest
```

```
mojorojo@mojorojo-vmware:~/nginxswarm$ docker service ps nginx-service
ID             NAME             IMAGE          NODE            DESIRED STATE   CURRENT STATE               ERROR     PORTS
kfkz52idnhej   nginx-service.1  nginx:latest   mojorojo-vmware Running         Running about a minute ago
ibrvoj0ecou2   nginx-service.2  nginx:latest   mojorojo-vmware Running         Running about a minute ago
mojorojo@mojorojo-vmware:~/nginxswarm$
```

Q5. Docker Swarm: Python Worker Service

Question

Deploy a Python script as a Swarm service.

Requirements

1. Create app.py: import time while True: print("Running in Swarm...") time.sleep(5)

```
mojorojo@mojorojo-vmware:~/pythonswarm$ cat app.py
import time
while True:
    print("Running in Swarm...")
    time.sleep(5)
mojorojo@mojorojo-vmware:~/pythonswarm$
```

2. Create Docker image

```
mojorojo@mojorojo-vmware:~/pythonswarm$ docker build -t nginx .
[+] Building 1.2s (8/8) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 133B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.11-alpine3.22                                                 1.0s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [1/3] FROM docker.io/library/python:3.11-alpine3.22@sha256:a4fc589b32e824f3f02ed9d7e7be19518aa47e105b80416336af9f202275a489  0.0s
 => => resolve docker.io/library/python:3.11-alpine3.22@sha256:a4fc589b32e824f3f02ed9d7e7be19518aa47e105b80416336af9f202275a489  0.0s
 => [internal] load build context                                                                                         0.0s
 => => transferring context: 27B                                                                                          0.0s
 => CACHED [2/3] WORKDIR /PYTHON-APP                                                                                      0.0s
 => CACHED [3/3] COPY APP.PY /PYTHON-APP                                                                                  0.0s
 => exporting to image                                                                                                    0.1s
 => => exporting layers                                                                                                   0.0s
 => => exporting manifest sha256:62c695ceed752ff803df154a5eb1817a567895edb729ed54faf9afa3bdcf272a                         0.0s
 => => exporting config sha256:2db0fce371a728ee4c85be7ccb5164988c81f152f215ad7f6a46a03a44c94f31                           0.0s
 => => exporting attestation manifest sha256:06369e39fd124205d51a2bfeed6a757fab43ca3f0dc169696b8012c7a5375a1b             0.0s
 => => exporting manifest list sha256:07970dac3f2466efc05bf19171cac99e6e98dc9a2dcacf53c994ac5a66e02e8f                    0.0s
 => => naming to docker.io/library/nginx:latest                                                                           0.0s
 => => unpacking to docker.io/library/nginx:latest                                                                        0.0s
```

3. Deploy service:
   - Name python-service
   - Replicas 3

```
mojorojo@mojorojo-vmware:~/pythonswarm$ docker service create --name python-service --replicas 3 nginx
iah7c0x0ad8o0r3g0bgpgzli4
overall progress: 3 out of 3 tasks
1/3: running   [==================================================>]
2/3: running   [==================================================>]
3/3: running   [==================================================>]
verify: Service iah7c0x0ad8o0r3g0bgpgzli4 converged
```

4. Check logs from different containers

```
mojorojo@mojorojo-vmware:~/pythonswarm$ docker service ls
ID             NAME             MODE         REPLICAS   IMAGE          PORTS
iah7c0x0ad8o   python-service   replicated   3/3        nginx:latest
```

```
mojorojo@mojorojo-vmware:~/pythonswarm$ docker service ps python-service
ID             NAME               IMAGE          NODE             DESIRED STATE   CURRENT STATE           ERROR     PORTS
qar19skeuz48   python-service.1   nginx:latest   mojorojo-vmware  Running         Running 7 minutes ago
tne54c2xk5e0   python-service.2   nginx:latest   mojorojo-vmware  Running         Running 7 minutes ago
jyt8jr5nd6yx   python-service.3   nginx:latest   mojorojo-vmware  Running         Running 7 minutes ago
mojorojo@mojorojo-vmware:~/pythonswarm$
```

Q.6 Python + Python (Multiple Services)

Create a Docker Compose setup with two Python containers:

- service1
- service2

Each container should run a different Python script and print different messages.

```
mojorojo@mojorojo-vmware:~/differentpythoncontainers$ cat docker-compose.yml
services:
  python_container:
    build: ./cont1
    container_name: python-cont
  python_container2:
    build: ./cont2
    container_name: python-cont2
mojorojo@mojorojo-vmware:~/differentpythoncontainers$
```

```
mojorojo@mojorojo-vmware:~/differentpythoncontainers/cont1$ cat Dockerfile
FROM    python:3.13-slim
WORKDIR /python-app
COPY    app1.py /python-app
CMD     ["python","app1.py"]
```

```
mojorojo@mojorojo-vmware:~/differentpythoncontainers/cont2$ cat Dockerfile
FROM    python:3.13-slim
WORKDIR /python-app2
COPY    app2.py /python-app2
CMD     ["python","app2.py"]
```

```
mojorojo@mojorojo-vmware:~/differentpythoncontainers$ docker compose up
[+] Building 8.2s (17/17) FINISHED
 => [internal] load local bake definitions                                                                    0.0s
 => => reading from stdin 1.16kB                                                                              0.0s
 => [python_container2 internal] load build definition from Dockerfile                                        0.0s
 => => transferring dockerfile: 132B                                                                          0.0s
 => [python_container internal] load build definition from Dockerfile                                         0.0s
 => => transferring dockerfile: 129B                                                                          0.0s
 => [python_container2 internal] load metadata for docker.io/library/python:3.13-slim                         2.4s
 => [python_container2 internal] load .dockerignore                                                           0.0s
 => => transferring context: 2B                                                                               0.0s
 => [python_container internal] load .dockerignore                                                            0.0s
 => => transferring context: 2B                                                                               0.0s
 => [python_container 1/3] FROM docker.io/library/python:3.13-slim@sha256:40365323acc4621904ca8d85aaf1a32da1ae2b93ecf98  4.8s
 => => resolve docker.io/library/python:3.13-slim@sha256:40365323acc4621904ca8d85aaf1a32da1ae2b93ecf98bbefe956601d1b57f  0.0s
 => => sha256:eb7aeb6e31180144e93bad8e32441038f4c4520ec825039827c05fadcc01305f 249B / 249B                    0.7s
 => => sha256:a51d847b5d4902dac7726e34078eeb8e64ee8047b67ea02877acabcef191b07f 11.82MB / 11.82MB              1.4s
 => => sha256:3f977313cbf20d1fcbf9d25f28735d97b7bc691319bc43f8c539c8d05c9cf886 1.29MB / 1.29MB                1.3s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB              2.5s
 => => extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd                     1.3s
 => => extracting sha256:3f977313cbf20d1fcbf9d25f28735d97b7bc691319bc43f8c539c8d05c9cf886                     0.1s
 => => extracting sha256:a51d847b5d4902dac7726e34078eeb8e64ee8047b67ea02877acabcef191b07f                     0.7s
 => => extracting sha256:eb7aeb6e31180144e93bad8e32441038f4c4520ec825039827c05fadcc01305f                     0.0s
 => [python_container internal] load build context                                                            0.0s
 => => transferring context: 66B                                                                              0.0s
 => [python_container2 internal] load build context                                                           0.0s
 => => transferring context: 66B                                                                              0.0s
 => [python_container 2/3] WORKDIR /PYTHON-APP                                                                0.3s
```

```
 => [python_container2 3/3] COPY APP2.PY /python-app2                                                         0.0s
 => [python_container] exporting to image                                                                     0.3s
 => => exporting layers                                                                                       0.1s
 => => exporting manifest sha256:38ad68bf46b89f76fcc5a527bee2bb6c9a82530dab43ce4481f3167dcead1ce4             0.0s
 => => exporting config sha256:91416843f0e3945ee380c6e082b641db91ec507483acfbf7bee6254eb1a3a86d               0.0s
 => => exporting attestation manifest sha256:daab478e4c34314912cc4bc845522283a135458bc28f3f3d85a8c735a9b1c910  0.0s
 => => exporting manifest list sha256:f91dc54bcef874d1ce64cab523b323dea3bba81c3351e99677ef6eb9a98ff515        0.0s
 => => naming to docker.io/library/differentpythoncontainers-python_container:latest                          0.0s
 => => unpacking to docker.io/library/differentpythoncontainers-python_container:latest                       0.0s
 => [python_container2] exporting to image                                                                    0.3s
 => => exporting layers                                                                                       0.1s
 => => exporting manifest sha256:a6e7e8f4a3ec0f8d029c3f2e86f1bd086119b9417f69a10dd4f68eaf908893d4             0.0s
 => => exporting config sha256:a914b73188510b83f2ea1579bca07011dece2d6e6842ed3d312c8348d90e62fd               0.0s
 => => exporting attestation manifest sha256:3debc3f172edcbf483b464556cf0ed80f2ca416fa41f3954f958e839c20ec5f6  0.0s
 => => exporting manifest list sha256:6f400de7750ff1f20740698651c930d4ab712d552aa4d109d4cfae4f14cfd360        0.0s
 => => naming to docker.io/library/differentpythoncontainers-python_container2:latest                         0.0s
 => => unpacking to docker.io/library/differentpythoncontainers-python_container2:latest                      0.0s
 => [python_container] resolving provenance for metadata file                                                 0.0s
 => [python_container2] resolving provenance for metadata file                                                0.0s
[+] up 5/5
 ✓ Image differentpythoncontainers-python_container2 Built                                                    8.2s
 ✓ Image differentpythoncontainers-python_container  Built                                                    8.2s
 ✓ Network differentpythoncontainers_default         Created                                                  0.1s
 ✓ Container python-cont                             Created                                                  0.1s
 ✓ Container python-cont2                            Created                                                  0.1s
Attaching to python-cont, python-cont2
python-cont2  | Hello from Container 2
python-cont   | Hello from Container 1
python-cont2 exited with code 0
python-cont exited with code 0
```

Q7. MySQL + Backup Container

Create a Docker Compose setup with:

- one MySQL container
- one backup container

```
mojorojo@mojorojo-vmware:~/pythontwosrv$ vim docker-compose.yaml
mojorojo@mojorojo-vmware:~/pythontwosrv$ docker compose up
[+] Building 3.3s (14/14) FINISHED
 => [internal] load local bake definitions                                                                    0.0s
 => => reading from stdin 1.00kB                                                                              0.0s
 => [backup_worker internal] load build definition from Dockerfile                                            0.1s
 => => transferring dockerfile: 84B                                                                           0.0s
 => [db_server internal] load build definition from Dockerfile                                                0.1s
 => => transferring dockerfile: 52B                                                                           0.0s
 => [db_server internal] load metadata for docker.io/library/mysql:8.0                                        0.0s
 => [backup_worker internal] load metadata for docker.io/library/alpine:latest                                0.0s
 => [backup_worker internal] load .dockerignore                                                               0.0s
 => => transferring context: 2B                                                                               0.0s
 => [db_server internal] load .dockerignore                                                                   0.0s
 => => transferring context: 2B                                                                               0.0s
 => CACHED [backup_worker 1/2] FROM docker.io/library/alpine:latest@sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145  0.0s
 => => resolve docker.io/library/alpine:latest@sha256:5b10f432ef3da1b8d4c7eb6c487f2f5a8f096bc91145e68878dd4a5019afde11   0.0s
 => [db_server 1/1] FROM docker.io/library/mysql:8.0@sha256:4af1f8815716546f5b12410f7621f37f93db8dd11a184706ef59111930b  0.2s
 => => resolve docker.io/library/mysql:8.0@sha256:4af1f8815716546f5b12410f7621f37f93db8dd11a184706ef59111930b8c2ff       0.0s
 => [backup_worker 2/2] RUN apk add --no-cache bash                                                           2.7s
```

```
mojorojo@mojorojo-vmware:~/pythontwosrv$ ls
backup  backup_storage  database_storage  db  docker-compose.yaml
mojorojo@mojorojo-vmware:~/pythontwosrv$ cd db
mojorojo@mojorojo-vmware:~/pythontwosrv/db$ ls
Dockerfile
mojorojo@mojorojo-vmware:~/pythontwosrv/db$ cd ..
mojorojo@mojorojo-vmware:~/pythontwosrv$ ls
backup  backup_storage  database_storage  db  docker-compose.yaml
mojorojo@mojorojo-vmware:~/pythontwosrv$ cd database_storage/
mojorojo@mojorojo-vmware:~/pythontwosrv/database_storage$ ls
 auto.cnf        ca-key.pem       '#ib_16384_0.dblwr'   ibtmp1          mysql.ibd            public_key.pem    undo_001
 binlog.000001   ca.pem           '#ib_16384_1.dblwr'  '#innodb_redo'   mysql.sock           server-cert.pem   undo_002
 binlog.000002   client-cert.pem   ib_buffer_pool      '#innodb_temp'   performance_schema   server-key.pem
 binlog.index    client-key.pem    ibdata1              mysql           private_key.pem      sys
mojorojo@mojorojo-vmware:~/pythontwosrv/database_storage$ cd ..
mojorojo@mojorojo-vmware:~/pythontwosrv$ docker ps
CONTAINER ID   IMAGE                    COMMAND                  CREATED             STATUS             PORTS
NAMES
07ed202aeeb0   pythontwosrv-db_server   "docker-entrypoint.s…"   58 seconds ago      Up 57 seconds      3306/tcp, 33060/tcp
pythontwosrv-db_server-1
96bd42f104b0   mysql:8.0                "docker-entrypoint.s…"   28 minutes ago      Up 28 minutes      3306/tcp, 33060/tcp
pythontwosrv-database-1
27e20ab56006   nginx:latest             "/docker-entrypoint.…"   About an hour ago   Up About an hour   80/tcp
```

The backup container should access MySQL data using a shared volume.

Q8. Write a steps for kubernetes Installation (Minikube)

Step 1:

`curl  -LO https://storage-googleapis.com/minikube/release/latest/minikube/releases/latest/minikube-linux-amd64`

step 2:

```
sudo install minikube-linux-amd64 /usr/local/bin/minikube
```

Step 3:

```
minikube start --driver=docker
```

Step 4:

```
minikube kubectl -- get nodes
```

Step 5:

```
echo 'alias kubectl="minikube kubectl --"' >> ~/.bashrc
```

Step 6:

```
source ~/.bashrc
```